

Alzheimer's disease Detection and Classification using Deep Learning Algorithms

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Abstract— Alzheimer's disease is a significant and growing challenge in the field of healthcare, characterized as a progressive neurodegenerative disorder that primarily affects memory, thinking, and behavior. It is caused by complex brain changes that occur over time, including the buildup of amyloid plaques and tangled tau proteins., which disrupt communication between neurons and ultimately lead to cell death. The impact of Alzheimer's extends beyond the individual, profoundly affecting families and caregivers, as it often results in severe cognitive decline and a loss of independence. Recent estimates indicate that millions of people are living with Alzheimer's, resulting in significant healthcare expenses and emotional strain. Thus, the urgent need for accurate diagnosis, effective treatments, and supportive care systems is critical for improving the quality of life for those affected and advancing our understanding of this complex disease. Hence, the aim of this proposed study is to address a critical issue in Alzheimer's disease detection by implementing the VGG16 algorithm for feature extraction and EfficientNetB0 and Tiny-CNN for classification. The approach achieves an accuracy of 98.98% in detecting and classifying Alzheimer's disease, providing an effective solution for early diagnosis.

KEYWORDS—Alzheimer's Disease, VGG16, EfficientNetB0, Tiny-CNN

I. INTRODUCTION

Alzheimer's disease (AD) is one of the most prevalent neurodegenerative disorders, characterized by a gradual decline in cognitive abilities, memory impairment, and disruptions in daily activities. As global population ages, the prevalence of AD is rising, making early diagnosis crucial for managing the progression of the disease and improving patient outcomes. While there is no cure, early detection allows for timely intervention, treatment, and care planning, significantly improving the quality of life for affected individuals.

MRI (Magnetic Resonance Imaging) scans have become a critical tool in the diagnosis and analysis of Alzheimer's disease due to their ability to capture detailed images of brain structures. By identifying structural changes, such as brain atrophy in specific regions like the hippocampus, MRI scans can provide valuable insights into the progression of Alzheimer's. However, manually analyzing MRI scans for subtle patterns associated with different stages of the disease can be time-consuming and prone to human error. This has led to the increasing use of artificial intelligence and deep learning techniques for automated detection and classification.

In this study, we propose a deep learning-based approach to improve the automatic identification and classification of Alzheimer's disease stages through MRI scans. The VGG16

convolutional neural network (CNN) is utilized for the detection phase, specifically for extracting key features from MRI images. VGG16 is a well-established deep learning model known for its ability to capture complex image features, making it suitable for detecting subtle changes in brain structures.

Following the detection phase, we implement the EfficientNetB0 CNN model and Tiny-CNN Model for classification. EfficientNetB0, known for its efficient scaling and high accuracy, is used to classify MRI images into four distinct categories: non-Alzheimer's (healthy control), very mild dementia, mild dementia, and moderate dementia. These classifications correspond to different stages of Alzheimer's disease, enabling a more detailed understanding of the patient's condition. The goal of this study is to provide a robust and accurate system for identifying Alzheimer's disease and classifying its severity, which can assist healthcare professionals in making informed decisions regarding patient care.

The proposed system leverages the strengths of both VGG16 and EfficientNetB0 to create a reliable method for detecting Alzheimer's disease and distinguishing between its stages, ultimately contributing to the advancement of early diagnosis and personalized treatment plans.

II. LITERATURE REVIEW

Ávila-Jiménez et al. (2024) developed a Deep Learning model to classify Alzheimer's disease (AD) in dementia patients, addressing the under-diagnosis of AD in older individuals. Their study proposed classification models based on AD risk factors and employed various rebalancing methods for dataset preprocessing. The model showed superior performance compared to established machine learning techniques, indicating its potential as an effective tool for assisting in AD diagnosis.

Houria et al. (2022) developed a multi-modality MRI fusion strategy for detecting Alzheimer's disease (AD) using deep learning. Their approach combined Diffusion Tensor Imaging (DTI) to assess white matter integrity with structural MRI (sMRI) to identify gray matter atrophy, important for early detection of mild cognitive impairment (MCI). Features were extracted using a 2-dimensional deep convolutional neural network (CNN) and classified with a support vector machine (SVM). This method effectively differentiates between AD, MCI, and cognitively normal individuals, highlighting the benefits of integrating multiple imaging modalities to improve diagnostic accuracy in AD.

Al Shehri et al. (2022) explored deep learning techniques for diagnosing and classifying Alzheimer's disease, addressing the challenges of manual diagnosis, which can be error-prone and time-consuming. Their model utilized advanced CNN architectures, specifically DenseNet-169 and ResNet-50, to classify stages of Alzheimer's disease. Findings showed that DenseNet-169 outperforms ResNet-50, emphasizing its

potential for real-time analysis and enhanced diagnostic accuracy in clinical settings.

Helaly et al. (2022) developed a framework for early detection of Alzheimer's disease (AD) using deep learning techniques. They highlighted the significance of early diagnosis due to the lack of effective cures for AD. The study employed convolutional neural networks (CNN) to classify four stages of the disease, utilizing both simple architectures and transfer learning with VGG19. In response to COVID-19, they proposed a web application for remote AD assessments to assist doctors and patients. The findings indicated that the CNN architectures reduced computational complexity while improving classification performance.

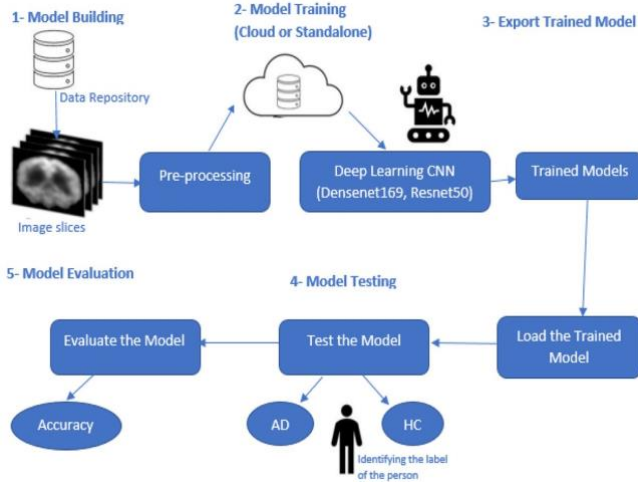
Raees et al. (2021) proposed an automated deep learning system for early detection of Alzheimer's disease (AD) using MRI data. The study underscores the need for early diagnosis due to the progressive and incurable nature of AD. They classified a dataset into Mild Cognitive Impairment (MCI), Alzheimer's disease, and Normal classes. Various classification methods, including Support Vector Machines (SVM) and Deep Neural Network (DNN) models, were tested, showing promising accuracy in predicting AD. The findings suggest that advanced machine learning tools can significantly improve early diagnosis of neurodegenerative diseases, enhancing patient outcomes.

Salehi et al. (2020) examined deep learning techniques for diagnosing Alzheimer's disease, emphasizing their advantages over traditional algorithms, particularly with high-dimensional MRI images. The article reviews various methods, highlighting Convolutional Neural Networks (CNN) as a common choice for brain image analysis. The authors concluded that early-stage prediction of Alzheimer's can be improved by applying advanced deep learning techniques across diverse datasets like ADNI and OASIS. This approach offers a promising pathway for enhancing diagnostic accuracy in Alzheimer's disease.

Ji et al. (2019) explored the early diagnosis of Alzheimer's disease (AD) using deep learning techniques, specifically convolutional neural networks (ConvNets), applied to magnetic resonance imaging (MRI). They highlighted the critical need for early confirmation to facilitate timely treatment and reduce brain damage. The study used MRI slices of gray and white matter for classification and employed ensemble learning to enhance accuracy by combining outputs from multiple classifiers. Three base ConvNets were designed and evaluated with a dataset from the Alzheimer's Disease Neuroimaging Initiative. The findings demonstrated promising accuracy in classifying AD and mild cognitive impairment, illustrating the potential of deep learning for early diagnosis.

III. METHODOLOGY

In this study, we focusing on identifying and categorizing Alzheimer's disease. into four distinct stages using advanced deep learning techniques, specifically leveraging the VGG16 and EfficientNetB0 convolutional neural networks (CNNs). Our dataset comprises 6400 MRI images, with 3200 images representing non-Alzheimer's cases and the remaining 3200 corresponding to different stages of Alzheimer's disease, classified as very mild dementia, mild dementia, and moderate dementia. This balanced dataset is essential for ensuring that the model can learn effectively from a variety of examples, facilitating accurate division.



To prepare the dataset for training, we split it into training and validation sets in an 80:20 ratio. This means 5,120 images are used for training the model, while the remaining 1,280 images are reserved for validation. This division is essential for evaluating how well the model generalizes its predictions to new, unseen data.

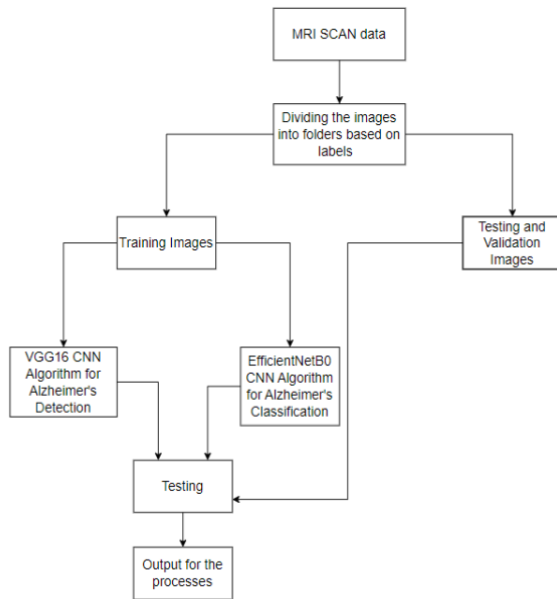


Fig1. Proposed Block Diagram

The methodology begins with employing the VGG16 CNN for the detection of Alzheimer's disease in MRI images. VGG16 is a well-established architecture known for its depth and capability to capture complex patterns in images. In our approach, we utilize VGG16 in its pre-trained form, taking advantage of its ability to recognize essential features from the MRI images without requiring any pre-processing steps such as resizing, normalization, or noise reduction. The model's architecture includes several convolutional layers followed by fully connected layers, which enable it to detect critical patterns associated with Alzheimer's disease. By utilizing the model in this way, we focus on accurately detecting the presence of Alzheimer's pathology in the MRI images.

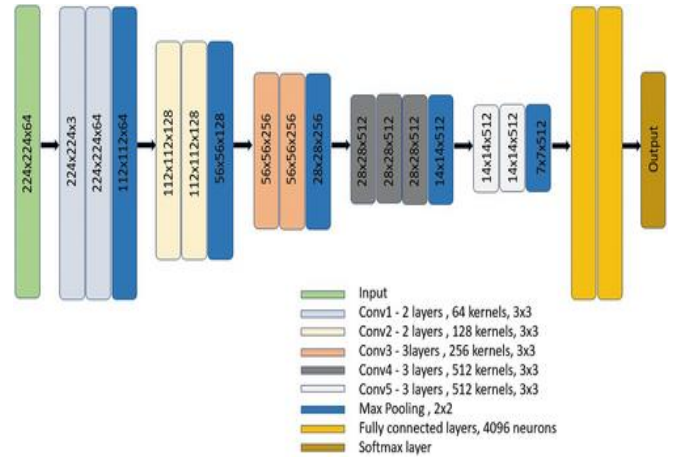


Fig2. VGG16 Architecture

Fig2. Represents the architecture of VGG16 CNN.

VGG16 is a deep learning model with 16 trainable layers, mainly composed of convolutional and fully connected layers. It starts with five blocks of convolutional layers, each followed by ReLU activation and max pooling. The convolutional layers use small 3×3 filters to capture detailed features. Following the convolutional blocks, VGG16 includes three fully connected layers. The first two contain 4,096 neurons each, while the final layer has 1,000 neurons for classification. To reduce overfitting, dropout is applied after the first two fully connected layers. The model's simple yet deep architecture makes it highly effective for image classification tasks. Once detection is performed using VGG16, we transition to the classification phase, where the EfficientNetB0 model is utilized. EfficientNetB0 is an advanced CNN architecture that excels in classification tasks due to its efficient scaling and optimization. By integrating the detected information from VGG16, EfficientNetB0 categorizes MRI images into four groups: non-Alzheimer's, very mild dementia, mild dementia, and moderate dementia. The model's efficiency and accuracy stem from its unique architecture, which balances depth, width, and resolution, making it well-suited for this classification challenge.

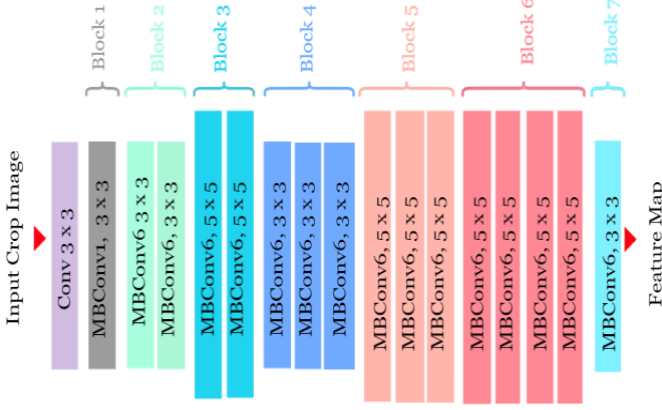


Fig3. EfficientNetB0 Architecture

Fig3. Represents the architecture of EfficientNetB0 CNN. EfficientNetB0 is a convolutional neural network (CNN) built for efficient scaling while achieving high accuracy. It features an input layer for 224x224 images, followed by a stem block with initial convolutions. The core consists of seven Mobile Inverted Bottleneck Convolutions (MBConv) with varying expansion ratios and output filters. Finally, it includes a 1x1 convolution, global average pooling, and a fully connected layer for classification. This architecture balances model size and performance, serving as the foundation for larger variants.

A Tiny CNN (Convolutional Neural Network) is a lightweight version of a traditional CNN, designed to be small, fast, and efficient while still capable of handling classification tasks. It consists of fewer convolutional layers, fewer filters per layer, and smaller kernel sizes (e.g., 3x3) to reduce computational complexity and memory usage. Tiny CNNs are especially useful for small datasets and low-power devices like smartphones and embedded systems. Despite their simplicity, they can still capture important spatial features through convolution and pooling operations. Typically, a Tiny CNN includes 2–4 convolutional layers followed by max-pooling layers to downsample the feature maps and reduce dimensions. After flattening the output, one or two fully connected layers are used for final classification. Activation functions like ReLU add non-linearity, and dropout layers are sometimes added to prevent overfitting. The output layer uses a softmax activation function for multi-class classification or a sigmoid function for binary classification. Tiny CNNs are fast to train and easy to deploy due to their low memory footprint. They are widely used in medical imaging, speech recognition, and real-time object detection where quick inference is essential.

During training, key parameters such as learning rate, batch size, number of epochs, and optimizer settings are carefully adjusted to enhance the model's performance. The categorical cross-entropy loss function is used to effectively handle the multi-class classification task, with accuracy being the primary evaluation metric.

Following the training phase, the model is validated using the 1280 images in the validation set. This step is vital for assessing how well the model can generalize to unseen data and ensuring its reliability in classifying MRI images accurately. The output of the model indicates whether an MRI image belongs to the non-Alzheimer's category or one of the Alzheimer's stages.

Overall, this methodology integrates the strengths of VGG16 and EfficientNetB0 and Tiny-CNN to provide a robust framework for the identification and classification of Alzheimer's disease. By prioritizing accurate detection and efficient classification, this approach seeks to improve diagnostic precision and support early intervention strategies in clinical practice.

IV. RESULTS AND DISCUSSIONS

In this proposed study of Alzheimer's detection and classification using VGG16 and EfficientNetB0 respectively, the source images are given and the output is predicted successfully with the trained model with the highest accuracy. The original image that is resized and for VGG16 algorithm it is known to take the RGB colored data as input for the model. However, MRI scan images are grayscale. So, it was necessary to make the grayscale data to multichannel before feeding it to VGG16.

VGG16 surprisingly had low performance considering that it is the most complicated model that I've used for the Alzheimer Detector.

This may be because of changing the greyscale to RGB (colored images) so that it fits the requirement for VGG16. (VGG16 is used for color images).

Thus, making the data change might have affected the performance of VGG16. Accuracy of somewhere around 50% just means that it is not different from making random guess as there are only two choices for the AI model to decide. 'This image is a Alzheimer patient's brain' or not.

For the classification of the images into four different categories, the used CNN algorithm-EfficientNetB0, was first trained with an Adam Optimizer with a learning rate of 0.1, then the accuracy obtained was around 50.7% and the loss was 6.202.

[6.201990604400635, 0.507031261920929]

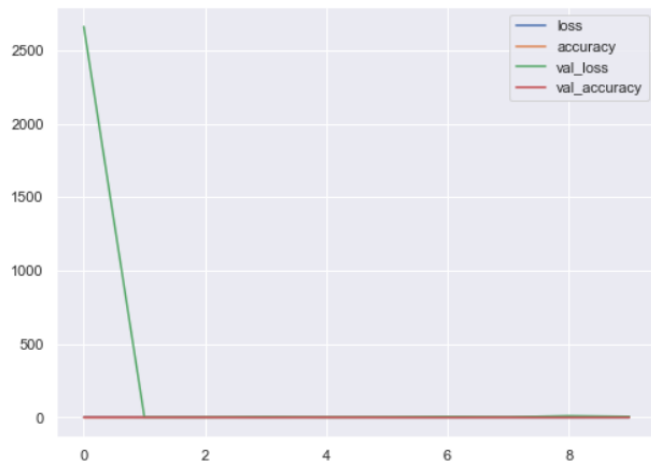


Fig.4. Loss and Accuracy for EffNetB0 model with LR0.1

The learning rate was further reduced to 0.01, then a different behavior of the model was observed. It exhibited a value loss of 0.4405 and accuracy of 85.46%.

[0.44051605463027954, 0.854687511920929]

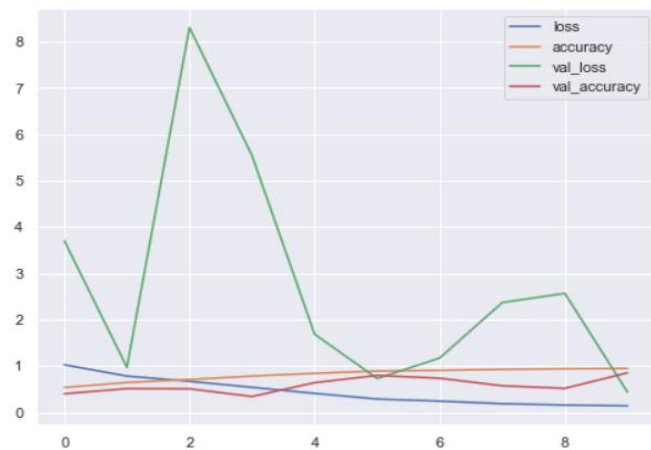


Fig.5. Loss and Accuracy for EffNetB0 model with LR0.01

EfficientNetB0 is considered to be a fine-tuned CNN algorithm, hence to further improve the model, resized images of 128x128 are fed rather than the images of size 224x224. Then, the loss obtained was 0.745 and the accuracy obtained was 83.52%.

[0.744975745677948, 0.835156261920929]

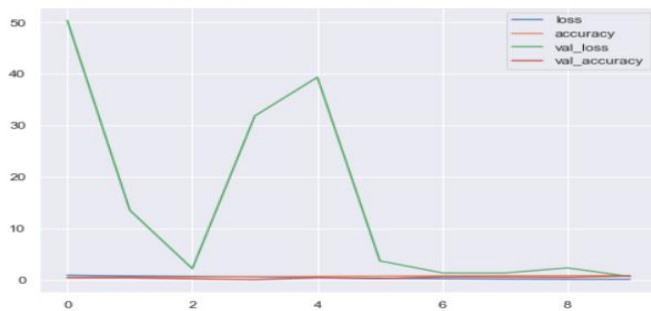


Fig.6. Loss and Accuracy for EffNetB0 model after resizing

In the process of using EfficientNetB0 for classifying the MRI images into four classes of Alzheimer's, different techniques were improvised out of which the technique with a learning rate of 0.01 had an accuracy of around 85%.

Further, the Tiny-CNN model was utilized for classifying the MRI Images into four classes of Alzheimer's. The accuracy is 98.98% being the highest of the two models used for classification.

Graphs comparing training and validation accuracy and loss are plotted to assess the model's performance.

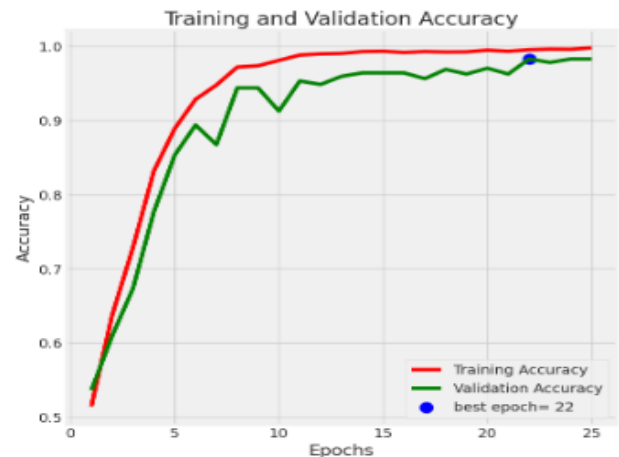


Fig.7. Accuracy for Tiny-CNN Model



Fig8. Loss for Tiny-CNN Model

The Training Vs Validation Accuracy graphs indicate that there is no overfitting in the model at all.

The performance of the model was further analyzed using a confusion matrix to compare the predicted classes against the true labels. The confusion matrix revealed that the model was particularly effective at identifying true positives (correctly identified cases of Alzheimer's Disease), with a relatively low number of false negatives, which is critical for medical

diagnoses where missing a case could have serious consequences.

True	Non-Demented	317	2	0	1
	Very Mild Dementia	2	315	0	3
	Mild Dementia	1	0	319	0
	Moderate Dementia	2	2	0	316
		Non-Demented	Very Mild Dementia	Mild Dementia	Moderate Dementia
		Predicted			

Fig9. Confusion Matrix

TABLE 1. Model Comparison

Proposed Method	Accuracy	Precision	Recall	F1-Score
Alzheimer's Detection using CNN(VGG16)	50%	0.72	0.56	0.63
Alzheimer's Disease Classification using CNN(EfficientNetB0)-LR0.1	50.7%	0.78	0.60	0.68
Alzheimer's Disease Classification using CNN(EfficientNetB0)-LR0.01	85%	0.83	0.88	0.86
Alzheimer's Disease Classification using CNN(EfficientNetB0)-Resize	83%	0.77	0.86	0.81
Tiny-CNN	98.98%	0.992	0.99	0.991

V. CONCLUSION

In conclusion, we adopted the VGG16 and EfficientNetB0 CNN and TinyCNN deep learning algorithms to accurately detect and classify the MRI scans into four different classes of Alzheimer's. The proposed methodology includes training and testing pothole images using pytorch, keras and tensorflow. Training data consisted of approximately 5120 images sourced from various locations. The efficacy of the model was evaluated and further the performance of the proposed model was further evaluated using various metrics and compared with existing models, as well as different parameter configurations of the same algorithm. The results indicate that the proposed model, based on TinyCNN, outperforms the existing approaches, achieving an accuracy of 98.98%. This demonstrates its potential as a promising solution for Alzheimer's classification.

REFERENCES

- [1] Ávila-Jiménez, J. L., Cantón-Habas, V., del Pilar Carrera-González, M., Rich-Ruiz, M., & Ventura, S. (2024). A deep learning model for Alzheimer's disease diagnosis based on patient clinical records. *Computers in Biology and Medicine*, 169, 107814.
- [2] Houria, L., Belkhamisa, N., Cherfa, A., & Cherfa, Y. (2022). Multi-modality MRI for Alzheimer's disease detection using deep learning. *Physical and Engineering Sciences in Medicine*, 45(4), 1043-1053.
- [3] Al Shehri, W. (2022). Alzheimer's disease diagnosis and classification using deep learning techniques. *PeerJ Computer Science*, 8, e1177.
- [4] Helaly, H. A., Badawy, M., & Haikal, A. Y. (2022). Deep learning approach for early detection of Alzheimer's disease. *Cognitive computation*, 14(5), 1711-1727.
- [5] Raees, P. M., & Thomas, V. (2021, May). Automated detection of Alzheimer's Disease using Deep Learning in MRI. In *Journal of Physics: Conference Series* (Vol. 1921, No. 1, p. 012024). IOP Publishing.
- [6] Salehi, A. W., Baglat, P., & Gupta, G. (2020). Alzheimer's disease diagnosis using deep learning techniques. *Int. J. Eng. Adv. Technol*, 9(3), 874-880
- [7] Liu, S., Liu, S., Cai, W., Pujol, S., Kikinis, R., & Feng, D. (2014, April). Early diagnosis of Alzheimer's disease with deep learning. In *2014 IEEE 11th international symposium on biomedical imaging (ISBI)* (pp. 1015-1018). IEEE.